

Samira Malek

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Summary

Ph.D. candidate in **Computer Science** specializing in **Machine Learning**, with 6+ years of experience in **deep learning, optimization, algorithms, and statistics**. Skilled in developing and deploying **large scale models** for AI systems and applications.

Education

Pennsylvania State University <i>Ph.D. in Computer Science & Engineering</i>	Pennsylvania, USA <i>2023–present</i>
Pennsylvania State University <i>M.S. in Computer Science & Engineering</i>	Pennsylvania, USA <i>2023–2025</i>
Sharif University of Technology <i>M.S. in Electrical Engineering</i>	Tehran, Iran <i>2018–2020</i>
Sharif University of Technology <i>B.S. in Electrical Engineering</i>	Tehran, Iran <i>2013–2018</i>

Research Interests

- Natural Language Processing
- Computational Imaging
- Communication
- Computer Vision
- Learning Theory
- Information Theory
- Optimization
- Explainable Deep Learning
- Signal Processing

Honors and Awards

- Awarded the College of Engineering Dean's Office scholarship at **Pennsylvania State University**. *2022*
- **Ranked 29th** in the Iran Nation-wide University Entrance Exam (Konkour) for M.S degree, among +40,000 test-takers. *2018*
- **Ranked 40th** in the Iran Nation-wide University Entrance Exam (Konkour) for B.S degree, among +84,000 test-takers. *2013*
- **Silver Medalist** of 30th Iran National Mathematical Olympiad. *2012*
- Admission to high school in **National Organization for Development of Exceptional Talents (NODET)** (success rate < 0.3%). *2009*

Experience

Pennsylvania State University <i>Research Assistant</i>	PA, USA <i>Jan 2023 – Present</i>
○ Designed and implement an interpretable, convergent convolutional neural network (CNN) for image deblurring achieving 32.84 dB PSNR/0.92 SSIM, and providing theoretical convergence proofs. ○ Evaluated advanced large language models (DeepSeek, Gemini 2.0, O-series) on Olympiad mathematics. ○ Created a multimodal dataset (1,135 questions) for discrete mathematics reasoning; benchmarked GPT-5, Gemini, Gemma. ○ Developed an agentic AI system for automated grading of mathematical proofs. ○ Skills: (non)convex optimization, torchvision, Agentic AI, designing and training DNN, Image Processing, Signal Processing, training on distributed computing, data collection	
Applied Research Laboratory, Pennsylvania State University <i>Machine Learning Researcher</i>	PA, USA <i>May 2024 – May 2025</i>
○ Built a hierarchical topic modeling using BERTopic + GPT, and generating sentence-level descriptions for topics. ○ Developed an LLM-based web application that detects health misinformation and generates warning message with 80% accuracy. ○ Skills: Finetuning generative language models, topic modeling, Prompt engineering, unsupervised clustering, Hugging face, OpenAI API	
SNOWA <i>Data & Analytics Engineer</i>	Tehran, Iran <i>May 2022–Dec 2022</i>
○ Developed customer segmentation using unsupervised learning to support targeted marketing. ○ Applied graph-based algorithms with SQL to identify and analyze customer relationship networks. ○ Designed and developed a scalable Data Warehouse to support business intelligence and analytics ○ Skills: SQL, Python, PCA, Microsoft Power BI, Microsoft SQL Server Management Studio, Scikit-learn, large-scale data engineering	

- Designed a **deep neural network** based on a **message-passing algorithm** for decoding linear codes, achieving BER of 10^{-7} at 6 dB SNR.
- Skills:** TensorFlow, NumPy, Pandas, Linux, Statistics, MAP Estimation, information theory, communication, graph-based models, MATLAB

Grants & Funding

- Financial assistance award from the Economic Development Administration Build Back Better Regional Challenge, Farms Food Future (F3) Innovation.
- Partial support from NSF and AgAID Institute (Agricultural AI for Transforming Workforce and Decision Support) under the USDA-NIFA award No. 2021-67021-35344.
- Research grant from the Investigator Initiated Studies Program of Merck Sharp & Dohme Corp (MISP #102050).

Publications

- Samira Malek**, Christopher Griffin, Robert D Fraleigh, Robert Lennon, Vishal Monga, Lijiang Shen, **"Intervention in Health Misinformation Using Large Language Models for Automated Detection, Thematic Analysis, and Inoculation: Case Study on COVID-19"**, Journal of Medical Internet Research, 2026.
- Hamed Mahdavi, Alireza Hashemi, Majid Daliri, Pegah Mohammadipour, Alireza Farhadi, **Samira Malek**, et al. **"Brains vs. bytes: Evaluating llm proficiency in Olympiad mathematics"**, Conference On Language Modeling, 2025.
- Samira Malek**, Christopher Griffin, Robert D Fraleigh, Robert Lennon, Vishal Monga, Lijiang Shen, **"A Methodology Framework for Analyzing Health Misinformation to Develop Inoculation Intervention Using Large Language Models: A case study on covid-19"**, AEJMC 108th Annual Conference, 2025.
- Hamed Mahdavi, Pouria Mahdavinia, Alireza Farhadi, Pegah Mohammadipour, **Samira Malek**, et al. **"CombiGraph-Vis: A Curated Multimodal Olympiad Benchmark for Discrete Mathematical Reasoning"**, The 5th Workshop on Mathematical Reasoning and AI at NeurIPS 2025.
- Hamed Mahdavi, Pouria Mahdavinia, **Samira Malek**, et al. **"RefGrader: Automated Grading of Mathematical Competition Proofs using Agentic Workflows"**, The 5th Workshop on Mathematical Reasoning and AI at NeurIPS 2025.
- Mohsen Farajijalal, **Samira Malek**, et al., **"Data-Driven Model to Improve Mechanical Harvesters for Nut Trees"**, ASABE Annual International Meeting, 2024.
- Samira Malek**, et al., **"Multi Variable-layer Neural Networks for Decoding Linear Codes"**, Iran Workshop on Communication and Information Theory (IWCIT), IEEE, 2020.
- Samira Malek**, Vishal Monga, et al. **"Cascaded, Convergent Unrolled Deep Neural Networks for Blind Image Deconvolution"**, submitted to IEEE Transactions .
- Samira Malek**, Vishal Monga, et al. **"Leveraging Phase Information to Boost Unrolled Network Learning for Image Deblurring"**, to be submitted to ECCV.
- Samira Malek**, Christopher Griffin, Robert D Fraleigh, Robert Lennon, Lijiang Shen, **"A Large Language Model-Based Analyzing of HPV and COVID-19 Vaccines Misinformation on Social Media"**, submitted to Digital Health Journal.
- Samira Malek**, et al., **"A Deep Neural Network Architecture for Decoding Linear Codes Based on the Parity Check Matrix"**, (it's available [here](#)).
- Samira Malek**, et al., **"TextGen-BERT Classifier"**, (it's available [here](#)).

Teaching Experience

- TA for Discrete Mathematics** in Pennsylvania State University. *Spring & Fall 2023, Spring 2024*
- TA for Signal & System** in Sharif University of Technology. *Fall 2019*
- TA for Stochastic Random Process** in Sharif University of Technology. *Spring 2019*
- TA & MATLAB Teaching for Engineering Mathematics** in Sharif University of Technology. *Spring 2019*
- Design Mock Test** of the Iran Nation-wide University Entrance Exam (Konkour) for B.S degree in Kanoon Institution, which has the most participants in Iran (more than +20,000 students every year). *2017–2018*
- Teaching Mathematics** Farzanegan High school (NODET), Preparing students for Iran National Mathematical Olympiad. *2014–2015*

Selected Courses

- Deep Learning for NLP
- Convex Optimization
- Large Scale Machine Learning
- Non-convex Optimization
- Machine Learning Algs & Tools
- Distributed Optimization

- Pattern Recognition & ML
- Probability and Statistics
- Communication Systems
- Secure & Robust ML
- Stochastic Random Process
- Digital Signal Processing
- AI and Biological Computation
- Speech Processing
- Image Processing

Selected Academic Projects

Deep Learning

Fallacy Science Question Answering with LLMs

Fall 2024

- Built dataset of fallacy-based science questions; evaluated Llama, Gemini, and ChatGPT, improving model accuracy on reasoning by 60% through optimized prompts.

Identifying the Source Model of Generated Text

Fall 2024

- Generated datasets using five large language models and trained BERT classifiers to distinguish text by source model.

Multi-hop Question Answering with GPT

Spring 2023

- Trained GPT models via prompting for multi-hop reasoning on the HotPotQA benchmark.

Pitch Tracking in Noisy Speech

Spring 2019

- Implemented feedforward and recurrent neural networks for pitch estimation in noisy conditions.

EEG-based Typing Classification

Spring 2016

- Designed a neural network to classify right- vs. left-handed typing using EEG signals.

Backdoor Attack and Detection on CIFAR-10

Spring 2024

- Designed and evaluated additive backdoor attacks and implemented detection methods for image classifiers.

Optimization

MinMax Optimization Algorithms

Spring 2023

- Implemented Stochastic Gradient Descent Ascent (SGDA) and Stochastic Compositional Gradient (SCSC) for min-max optimization problems.

Classical Optimization Methods

Fall 2018

- Implemented BFGS, Steepest Descent, and Newton's method with line-search for nonlinear optimization.

Signal Processing & Communications

Viterbi Algorithm and Sequence Detection

Spring 2020

- Implemented the Viterbi algorithm and developed a reduced-complexity sequence detector for communication systems.

Pitch Contour Extraction

Spring 2019

- Extracted pitch contours from speech signals using SIFT, HPS, and AMDF algorithms.

Image Recovery via Sparse Methods

Fall 2018

- Applied IMAT and OMP algorithms for image recovery from incomplete data.

Signal Reconstruction Techniques

Fall 2018

- Reconstructed 1-D and 2-D signals using SDFT and RS methods.

Computer Skills

Programming: Python, MATLAB, SQL, C, Java, R, HTML, Git, Linux, GPU/CUDA

Tools: PyTorch, Tensorflow, Pandas, OpenCV, scikit-learn, Hugging Face, OpenAI API, Transformers, torchvision, NumPy, Pandas, Linux, Git, AWS, Prompt Engineering, Training Generative Models

Languages

- English
- Persian
- Azerbaijani